The Complete Statistics Tutorial

From Descriptive Statistics to Hypothesis Testing and the T-Test

A step-by-step written guide covering what statistics is, descriptive statistics, inferential statistics, levels of measurement, and the t-test — with worked examples and visualizations for every concept.

PART 1 Foundations of Statistics

PART 2 Hypothesis Testing & the T-Test

PART 3 Levels of Measurement

Dark Mode Edition — reformatted for comfortable low-light reading.

PART 1

Foundations of Statistics

What statistics is, descriptive statistics, and inferential statistics

1.1 What Is Statistics?

Statistics deals with the collection, analysis, and presentation of data. It gives us tools to turn raw observations into information we can actually use and understand.

EXAMPLE — Gender and newspaper preference

Suppose we want to investigate whether gender has an influence on the preferred newspaper. Gender and newspaper are the two variables we want to analyze. To study this, we first collect data — for instance, by creating a questionnaire that asks people their gender and their preferred newspaper, sending it out, and waiting for responses.

The collected answers can be displayed in a table. Each column represents one variable (gender, newspaper), and each row represents the response of one surveyed person.

Respondent	Gender	Preferred Newspaper
1	Male	New York Post
2	Female	USA Today
3	Male	New York Post
...	...	...

A simple data table: one row per respondent, one column per variable.

Data does not have to come from a survey — it can just as easily come from an experiment, such as testing the effect of two different drugs on blood pressure.

Once data is collected, the next question is: what do we actually want to do with it? We rarely survey an entire population — we usually only take a sample. This leads to two possible goals:

- If we only want to **describe the sample** we collected, we use **descriptive statistics**.
- If we want to **draw conclusions about the whole population** based on the sample, we use **inferential statistics**.

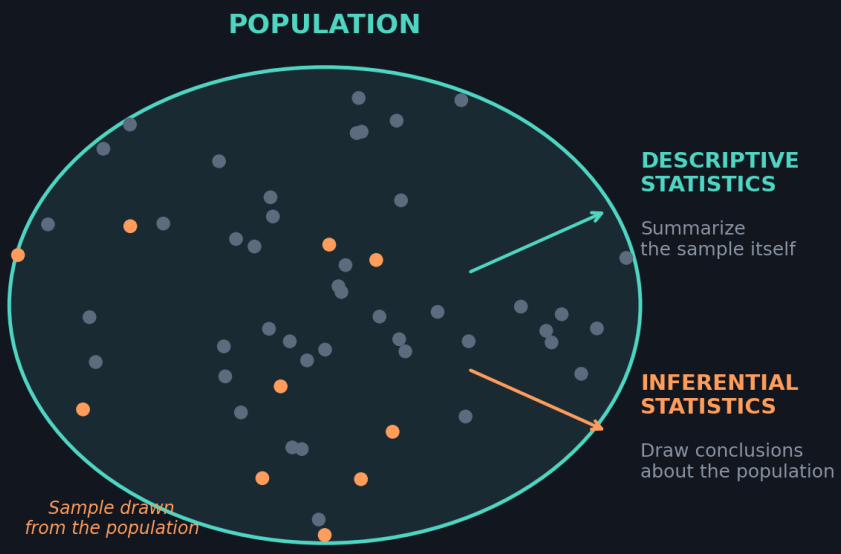


Figure 1.1 — A sample is drawn from the population. Descriptive statistics summarizes the sample; inferential statistics uses the sample to draw conclusions about the population.

✓ Key takeaway

- Statistics = collecting, analyzing, and presenting data.
- Descriptive statistics describes only the data you collected.
- Inferential statistics uses a sample to make conclusions about a larger population.

1.2 Descriptive Statistics

Descriptive statistics aims to describe and summarize a data set in a meaningful way. It is important to remember that descriptive statistics only describes the collected data — it does not draw conclusions about a larger population.

> **EXAMPLE — Company commute survey**
>
> A company wants to know how its employees travel to work, so it creates a survey. Once enough responses are collected, the company can describe this data using descriptive statistics. But just because we know how employees at this company commute doesn't mean we can say how all working people commute — that would require inferential statistics.

To describe data, we rely on four key components:

- Measures of central tendency — mean, median, mode
- Measures of dispersion — variance, standard deviation, range, interquartile range
- Frequency tables and contingency tables
- Charts — bar charts, pie charts, histograms, box plots

1.2.1 Measures of Central Tendency

The Mean

The arithmetic mean is the sum of all observations divided by the number of observations.

$$\textit{mean} = \textit{(sum of all values)} \div \textit{(number of values)}$$

> **EXAMPLE — Mean test score**
>
> Imagine five students scored 78, 85, 90, 92, and 88 on a test. To find the mean, we sum the scores (78 + 85 + 90 + 92 + 88 = 433) and divide by the number of scores (5). The mean test score is 86.6.

The Median

When values are arranged in ascending order, the median is the middle value. If there is an odd number of data points, the median is simply the middle value. If there is an even number, the median is the average of the two middle values. Importantly, the median is resistant to outliers.

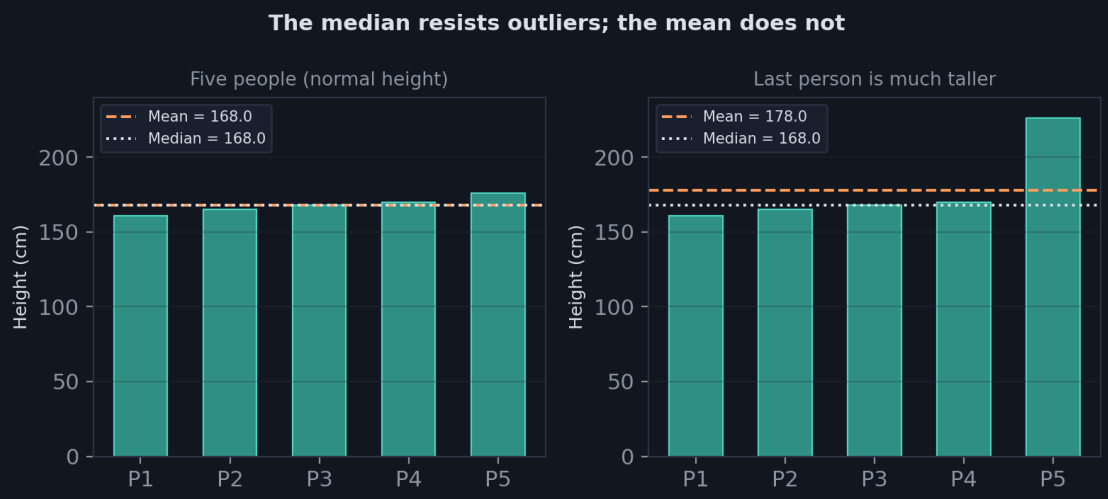


Figure 1.2 — No matter how tall the last person is, the person in the middle stays the middle — so the median doesn't move. The mean, however, gets pulled upward by the outlier.

The Mode

The mode is the value (or values) that appear most frequently in a data set.

EXAMPLE — Commuting mode

If 14 people travel to work by car, 6 by bike, 5 walk, and 5 take public transport, then car occurs most often — so car is the mode.

1.2.2 Measures of Dispersion

Measures of dispersion describe how spread out the values in a data set are. The main measures are the standard deviation, the variance, the range, and the interquartile range (IQR).

Standard Deviation

The standard deviation indicates the average distance between each data point and the mean. Every observation deviates from the mean by some amount; the standard deviation tells us, on average, how large that deviation is.

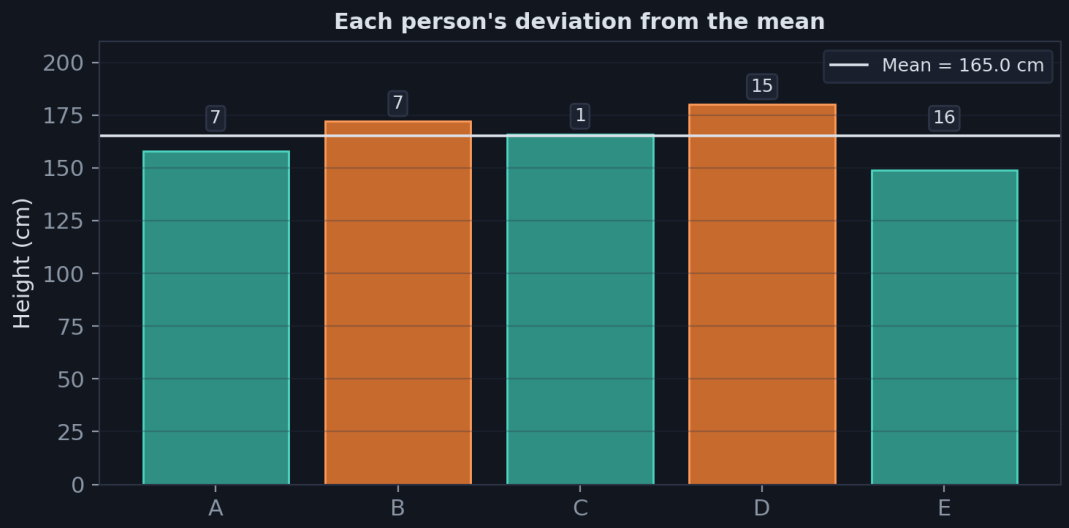


Figure 1.3 — Each person's height deviates from the group mean by a different amount. The standard deviation summarizes these deviations into a single average distance.

$$\sigma = \sqrt{[\ \Sigma(xi - M)^2 \div n\ ]}$$

Here σ is the standard deviation, n is the number of observations, xi is each individual value, and M (the mean, sometimes written “x-bar”) is the mean of all values.

There are two closely related versions of this formula: one divides by n, and the other divides by n – 1. A simple rule of thumb: if your survey or study does not cover the entire population (which is almost always the case), use the n – 1 version to estimate the population's standard deviation from your sample.

Variance

The variance is simply the squared standard deviation. Where the standard deviation is the quadratic mean of the distances from the mean, the variance is that same quantity before the square root is taken.

Range and Interquartile Range (IQR)

The range is the difference between the maximum and minimum value in the data set — simple, but sensitive to outliers.

The interquartile range (IQR) represents the middle 50% of the data. It is the difference between the first quartile (Q1) and the third quartile (Q3): 25% of values fall below Q1, 25% fall above Q3, and the IQR

itself contains exactly the middle 50% of values.

$$IQR = Q3 - Q1$$

✓ Central tendency vs. dispersion

- Measures of central tendency (mean, median, mode) give a single value that represents the whole data set.
- Measures of dispersion (standard deviation, range, IQR) show how spread out the data points are around that central value.
- Together, they tell you both 'where the data is centered' and 'how much it varies'.

1.2.3 Frequency Tables and Contingency Tables

A frequency table displays how often each distinct value appears in a data set.

EXAMPLE — Commute frequency table

A company surveyed 30 employees about how they get to work: car, bicycle, walk, or public transport. Counting each option gives a frequency table showing that car is the most common mode, chosen by 14 employees.

Mode of transport	Frequency	Percentage
Car	14	46.7%
Bicycle	6	20.0%
Walk	5	16.7%
Public transport	5	16.7%
Total	30	100%

But what if we have two categorical variables instead of one? This is where the contingency table (also called a cross-tab) comes in. It provides a way to analyze the relationship between two categorical variables: rows represent the categories of one variable, columns represent the categories of the other, and each cell shows the number of observations for that combination.

EXAMPLE — Two factory locations

Suppose the company has two factories: one in Detroit and one in Cleveland. We ask employees both how they commute and where they work. The resulting contingency table shows, for example, that car and Detroit were reported together 6 times.

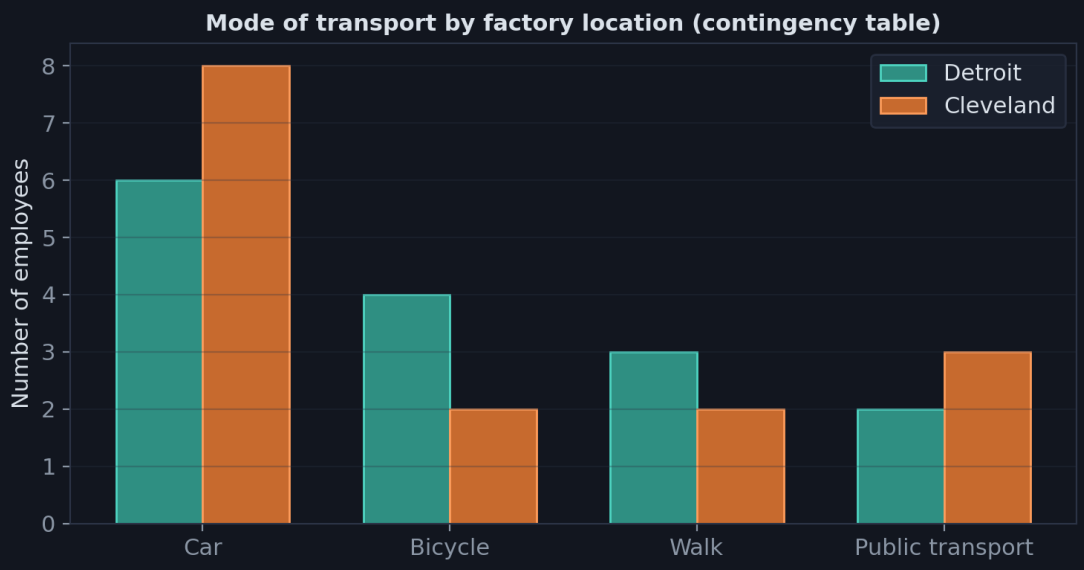


Figure 1.4 — A grouped bar chart visualizing a contingency table: mode of transport broken down by factory location.

1.2.4 Charts

Charts turn tables into visuals that are easier to grasp at a glance. The most common charts for the commute example are the bar chart and the pie chart.

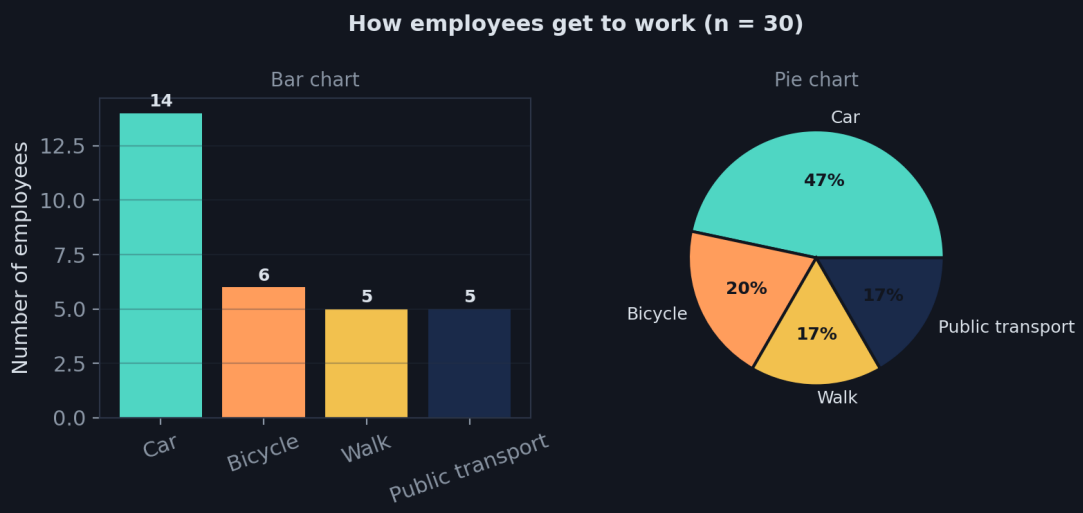


Figure 1.5 — The same frequency data shown as a bar chart (counts) and a pie chart (proportions).

Beyond bar and pie charts, other useful visualizations include the histogram (distribution of a metric variable), the box plot (median, quartiles, and outliers at a glance), the violin plot (a box plot combined with a distribution shape), and the rainbow / raincloud plot (raw data points plus distribution shape together). Which chart to use depends heavily on the level of measurement of your variable — a topic covered in detail in Part 3.

1.3 Inferential Statistics

Inferential statistics allows us to make a conclusion, or inference, about a population based on data from a sample.

- The **population** is the entire group we're interested in. Example: all adults in the United States.
- The **sample** is a smaller group actually studied, chosen from the population. Example: 150 adults selected from the United States.

We then use the sample to make a statement about the population. This process follows six steps.

Step 1 — Formulate a Hypothesis

First, we need a statement — a hypothesis — that we want to test.

> **EXAMPLE — Blood pressure drug**
>
> We want to know whether a drug has a positive effect on blood pressure in people with high blood pressure. Our population is therefore all people with high blood pressure (say, in the US). Since we cannot survey the entire population, we take a sample and use it to make a statement about the population as a whole.

Step 2 — Run a Hypothesis Test

Hypothesis testing is a method for testing a claim about a population parameter using data measured in a sample. Here's the logic behind it:

- We start with a **research hypothesis** (also called the **alternative hypothesis**) — the thing we're actually trying to find evidence for. Example: “the drug has an effect on blood pressure.”
- Because we cannot test this directly, we instead test the opposite: the **null hypothesis**, that the drug has *no* effect on blood pressure.
- We assume the null hypothesis is true — i.e., that people who take the drug and people who don't have the same average blood pressure.
- If a random sample then shows a large effect, we ask: *how likely is it to draw such a sample (or one that deviates even more) if the drug really has no effect?*
- If that probability is very low, we have evidence to **reject the null hypothesis**.

That probability is called the p-value. In three simple steps:

Step	What happens
1	The null hypothesis states there is no difference in the population.
2	The hypothesis test calculates how much the sample deviates from the null hypothesis.
3	The p-value gives the probability of getting a sample that deviates as much as (or more than) ours, assuming the null hypothesis is true.

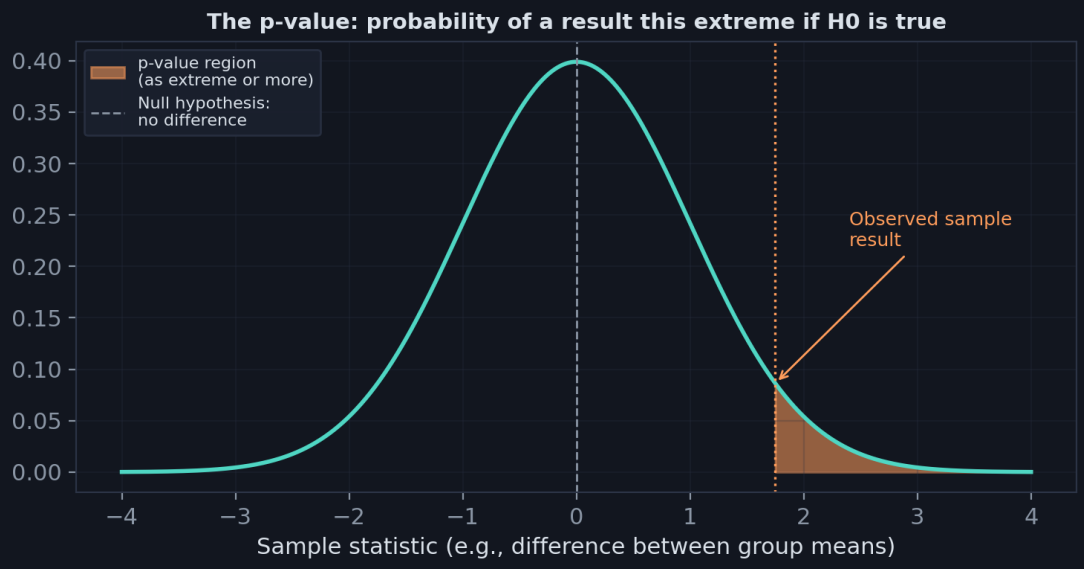


Figure 1.6 — The p-value is the probability, under the null hypothesis, of observing a result as extreme as (or more extreme than) the one actually measured in the sample.

Step 3 — Judge Statistical Significance

If the p-value is less than a predetermined threshold, the result is considered statistically significant — unlikely to have occurred by chance alone, and enough evidence to reject the null hypothesis. This threshold, the significance level, is most often set at 0.05 (5%).

- **Small p-value** (below the threshold): the data is inconsistent with the null hypothesis → reject the null hypothesis in favor of the alternative hypothesis.
- **Large p-value** (above the threshold): the data is consistent with the null hypothesis → we do not reject it.

Important caveat: a small p-value does not *prove* the alternative hypothesis is true — it only says such a result is unlikely if the null hypothesis were true. Likewise, a large p-value does not *prove* the null hypothesis is true — it only says such a result is likely under the null hypothesis. Statistical tests never offer proof, only evidence.

Step 4 — Understand Type I and Type II Errors

Because we're working with samples rather than the whole population, there is always a risk of making an error. There are two kinds:

- **Type I error** (false positive): a *true* null hypothesis is rejected. In the drug example: the drug actually has no effect, but our sample happened to look different enough that we mistakenly concluded it worked.
- **Type II error** (false negative): a *false* null hypothesis is *not* rejected. In the drug example: the drug actually does work, but our sample happened to show little difference, so we mistakenly concluded it didn't work.

	H0 is actually True	**H0 is actually False**
Reject H0	Type I Error (false positive) probability = alpha	Correct decision (true positive)
Do not reject H0	Correct decision (true negative)	Type II Error (false negative) probability = beta

Figure 1.7 — Type I and Type II errors, shown against the two possible true states of the world.

✓ Inferential statistics in a nutshell

- Population = everyone we care about. Sample = the smaller group we actually measured.
- Null hypothesis (H0): no effect / no difference. Alternative hypothesis (H1): there IS an effect.
- p-value < significance level (usually 0.05) → reject H0.
- Type I error = rejecting a true H0. Type II error = failing to reject a false H0.

Step 5 & 6 — Choose and Run the Right Test

The remaining steps are choosing an appropriate hypothesis test for your data and your question, and then calculating and interpreting it. Which test is appropriate depends heavily on your data's level of measurement, covered in Part 3, and on your specific research question. Part 2 walks through the most widely used test in introductory statistics: the t-test.

PART 2

Hypothesis Testing & The T-Test

One-sample, independent-samples, and paired-samples t-tests, step by step

2.1 What Is a T-Test?

The t-test is a statistical test procedure that analyzes whether there is a significant difference between the means of two groups.

> **EXAMPLE — Two drugs on blood pressure**
>
> Two groups of patients receive drug A and drug B, respectively. A t-test can tell us whether there is a meaningful difference in blood pressure between these two groups.

There are three types of t-tests, each suited to a different situation.



Figure 2.1 — The three t-test variants: comparing a sample to a fixed value, comparing two independent groups, and comparing the same group measured twice.

2.1.1 One-Sample T-Test

Used when we want to compare the mean of a sample with a known reference value.

> **EXAMPLE — Chocolate bar weight**
>
> A manufacturer claims its chocolate bars weigh 50g on average. We weigh a sample of 30 bars and get a mean of 48g. A one-sample t-test tells us whether 48g is significantly different from the claimed 50g, or whether this could just be normal sampling variation.

2.1.2 Independent-Samples T-Test

Used when we want to compare the means of two independent groups.

> **EXAMPLE — Comparing two painkillers**
>
> 60 people are randomly divided into two groups. Group 1 receives drug A; group 2 receives drug B. An independent-samples t-test tells us whether there is a significant difference in pain relief between the two drugs.

2.1.3 Paired-Samples T-Test

Used to compare the means of two dependent (paired) measurements — typically the same subjects measured twice.

> **EXAMPLE — Before-and-after diet**
>
> We weigh 30 people before a diet, then weigh the exact same people again after the diet. A paired-samples t-test checks whether the average weight difference before vs. after is statistically significant.

Interesting fact: a paired-samples t-test is mathematically very similar to a one-sample t-test. You can think of it as taking the difference for each pair (after minus before, for example) and then running a one-sample t-test on those differences against a reference value of zero — testing whether the average difference is significantly different from no change at all.

2.2 Assumptions of the T-Test

Every t-test requires:

- A suitable sample — a sample plus a reference value (one-sample), two independent samples (independent-samples), or one paired sample (paired-samples).
- The tested variable must be metric (e.g., age, body weight, income) — not something like education level.
- The metric variable should be approximately normally distributed in all three test variants.
- For the independent-samples t-test specifically, the variances in the two groups should be approximately equal (this can be checked with Levene's test).

2.3 Hypotheses of the T-Test

Test type	Null hypothesis (H0)	Alternative hypothesis (H1)
One-sample	Sample mean = reference value	Sample mean ≠ reference value
Independent-samples	Group means are equal	Group means are not equal
Paired-samples	Mean of the paired differences = 0	Mean of the paired differences ≠ 0

EXAMPLE — Why we need a t-test at all

Suppose we suspect a difference in bachelor's degree study duration between men and women in Germany. Our population is all bachelor's graduates in Germany, but we can't survey everyone, so we take a representative sample. Even if there is truly no difference in the population, a random sample will almost never show an exact difference of zero. The t-test tells us: at what measured difference can we confidently say the true difference is not zero?

2.4 Calculating the T-Test

To calculate a t-test, we first compute the t-value, which needs two ingredients: the difference between means, and the standard error (the standard deviation of that difference).

Test type	What's compared	Standard error uses
One-sample	Sample mean vs. reference value	Sample standard deviation and sample size (n)
Independent-samples	Mean of group 1 vs. mean of group 2	Standard deviations and sizes of both groups
Paired-samples	Mean of the paired differences vs. 0	Standard deviation of the differences, same as one-sample

t = (difference between means) ÷ (standard error of that difference)

- The t-value grows **larger** when the difference between means is larger.
- The t-value grows **smaller** when the difference between means is smaller.
- The t-value grows **smaller** when the data is more spread out (larger dispersion) — the more scattered the data, the less meaningful a given mean difference becomes.

2.5 From T-Value to a Decision: Two Equivalent Paths

Once we have a t-value, there are two equivalent ways to decide whether to reject the null hypothesis.

Path A — Compare to a Critical T-Value

We look up a critical t-value from a t-distribution table, based on our chosen significance level and the degrees of freedom.

- One-sample and paired-samples t-tests: degrees of freedom = n – 1.
- Independent-samples t-test: degrees of freedom = n1 + n2 – 2 (this can differ depending on whether equal or unequal variances are assumed).

EXAMPLE — Reading the table

With a 5% significance level and 9 degrees of freedom (a sample of 10 people), the critical t-value is 2.262. If our calculated t-value (say, 2.5) is greater than the critical value, we reject the null hypothesis — the means are different enough to be statistically significant.

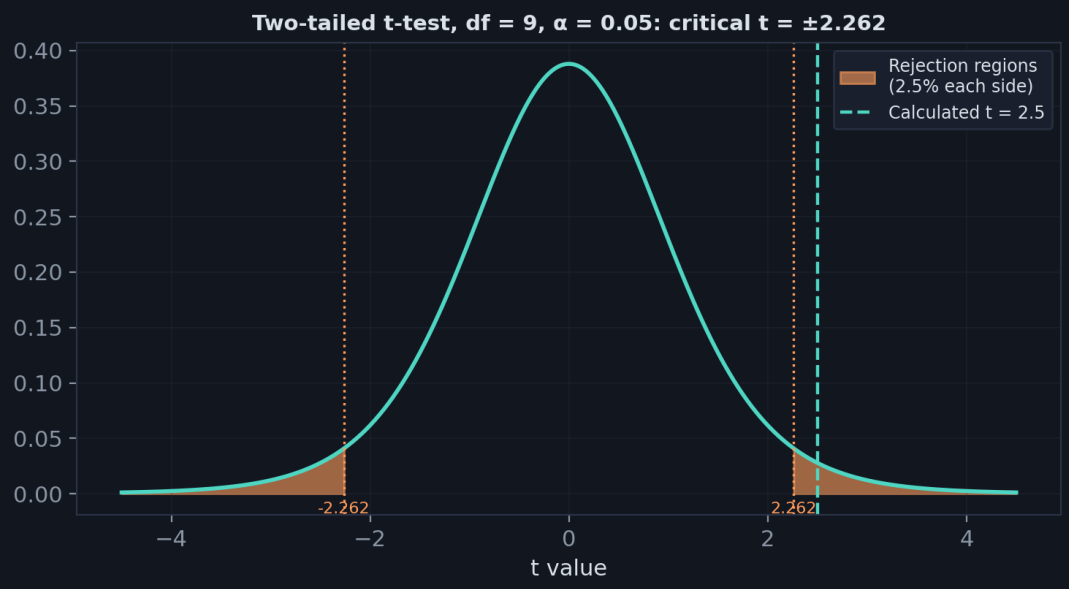


Figure 2.2 — The t-distribution with 9 degrees of freedom. If the calculated t-value falls in the shaded rejection region, we reject the null hypothesis.

Path B — Convert the T-Value into a P-Value

Alternatively, we can convert the t-value directly into a p-value.

EXAMPLE — From t to p

A t-value of 2.5 with 9 degrees of freedom converts to a p-value of 0.034. Since 0.034 is less than 0.05, we reject the null hypothesis — exactly the same conclusion as Path A. As a sanity check: a t-value of exactly 2.262 (the critical value itself) converts to a p-value of exactly 0.05, the boundary between significant and not significant.

✓ Two paths, same answer

- Calculated t-value > critical t-value ⇔ p-value < significance level.
- Both lead to the exact same decision about the null hypothesis.
- Software (like statistical calculators or R/Python/SPSS) typically reports the p-value directly.

2.6 Directed vs. Undirected Hypotheses (One-Tailed vs. Two-Tailed)

There's one more distinction worth understanding before wrapping up the t-test.

- **Undirected (two-tailed) hypothesis:** we only care whether there IS a difference, in either direction. Example: “there is a difference between the salary of men and women.” We don't care who earns more.
- **Directed (one-tailed) hypothesis:** we care about the direction of the difference. Example: “men earn more than women” or “women earn more than men.”

This changes where the rejection region sits on the t-distribution. In a two-tailed test at 5% significance, the 5% is split into two 2.5% regions, one on each tail. In a one-tailed test, the full 5% sits in a single tail, in the direction being tested.

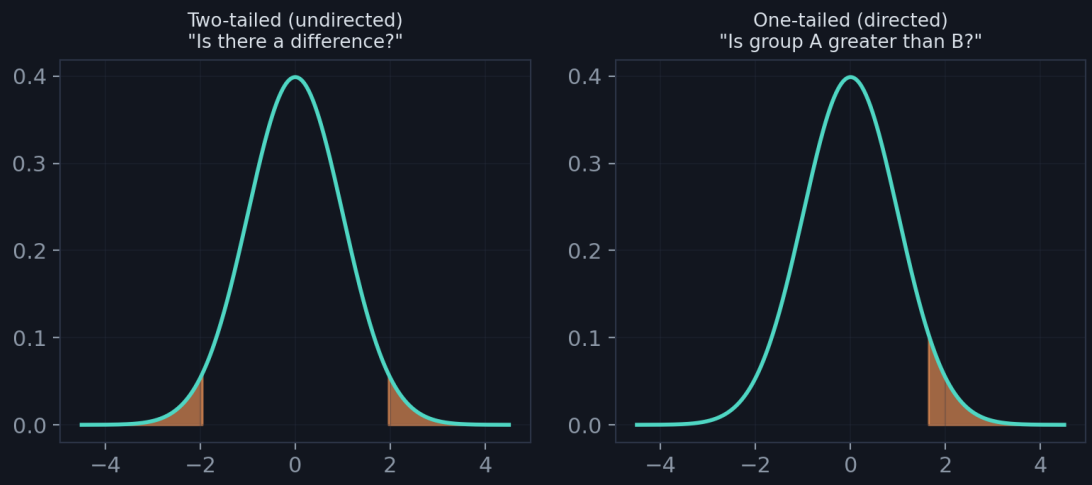


Figure 2.3 — Two-tailed tests split the rejection region across both ends of the distribution; one-tailed tests put the full rejection region on one side.

A t-test is a powerful tool for comparing the means of exactly two groups. When you need to compare more than two groups at once, the appropriate extension is analysis of variance (ANOVA) — a natural next step once the t-test feels comfortable.

PART 3

Levels of Measurement

Nominal, ordinal, interval, and ratio data — and why the distinction matters

3.1 What Are Levels of Measurement?

Levels of measurement refer to the different ways that variables can be quantified or categorized. Every variable in a data set corresponds to one of four primary levels: nominal, ordinal, interval, and ratio. In practice, interval and ratio data are often analyzed the same way, so together they're commonly called metric data.

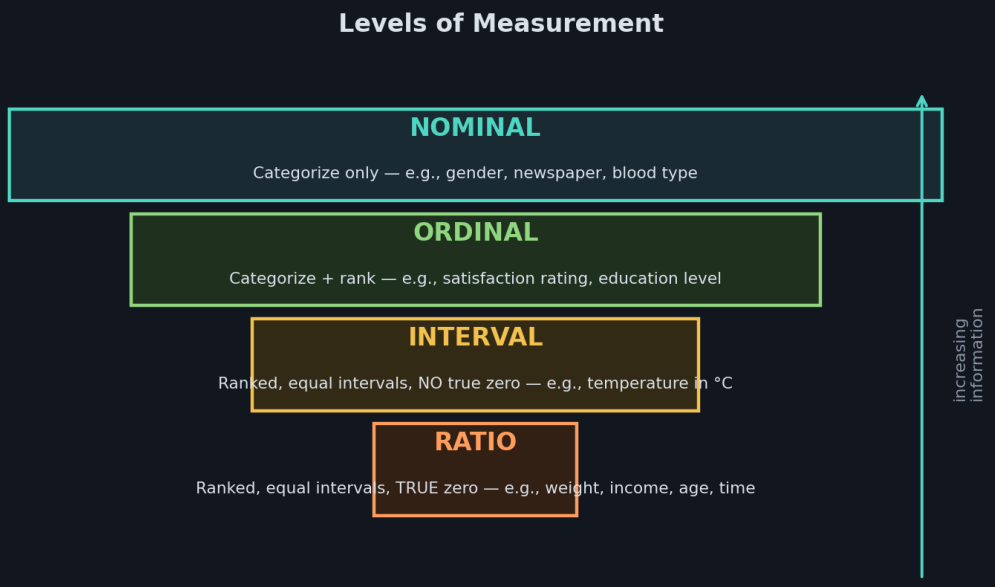


Figure 3.1 — The four levels of measurement, from least (nominal) to most (ratio) information content.

3.2 Why Levels of Measurement Matter

The level of measurement tells us how data can be collected, analyzed, and interpreted. Specifically, it determines:

- Which **statistical analyses** are valid — e.g., mean and standard deviation are suitable for metric data, sometimes usable for ordinal data (with care), and generally meaningless for nominal data.
- Which **hypothesis tests** are possible.
- Which **visualization** is most effective — bar charts suit nominal data well, while histograms suit metric data.

Crucially, the level of measurement matters mainly at the end of the research process (during analysis) — but it must be considered from the very start, when you decide what data to collect and how to form your questions. Choosing the wrong level of measurement at the start can make your desired analysis impossible later.

3.3 The Four Levels, One by One

3.3.1 Nominal

The most basic level. Nominal data can be categorized, but the categories cannot be meaningfully ranked.

> **EXAMPLE — Nominal variables**
>
> Gender (male/female), type of animal (dog/cat/bird), preferred newspaper. You can tell whether two values are the same or different, but you cannot say one category is “higher” or “lower” than another. In a questionnaire, the order in which nominal categories are listed usually doesn't matter. Nominal data is well suited to frequency tables and bar charts.

3.3.2 Ordinal

Ordinal data can be categorized and ranked in a meaningful order — but the differences between ranks are not mathematically meaningful (the intervals aren't necessarily equal).

> **EXAMPLE — Ordinal variables**
>
> Rankings (1st, 2nd, 3rd), satisfaction ratings (very unsatisfied, unsatisfied, neutral, satisfied, very satisfied), and education level (high school, bachelor's, master's). “How satisfied are you with your current job?” with those five response options is a classic ordinal question — there's a clear order, but the emotional “distance” between “neutral” and “satisfied” isn't necessarily the same as between “satisfied” and “very satisfied.”

3.3.3 Interval and Ratio (Metric)

Metric variables are the highest level of measurement: like ordinal data but with the added property that intervals between values are equally spaced, so differences and sums are mathematically meaningful.

> **EXAMPLE — Metric variables**
>
> Income, weight, age, and electricity consumption. In a questionnaire, a metric variable is usually just a direct input field — e.g., you type your exact age or body weight.

Metric data splits further into interval and ratio scales. The distinguishing feature is whether there is a true zero point.

EXAMPLE — Marathon times: ratio vs. interval

In a marathon, if the stopwatch starts at zero when the race begins, we have a true zero — every runner starts from nothing. This means we can say the fastest runner (2 hours) was three times as fast as the slowest (6 hours). This is ratio level.

But if the stopwatch was forgotten and only differences relative to the fastest runner were recorded, there's no true zero. We can still say meaningful things like “Runner B finished 1 hour after the fastest runner,” but we can no longer say the fastest runner was three times as fast as the slowest — we don't know the runners' absolute total times. This is interval level.

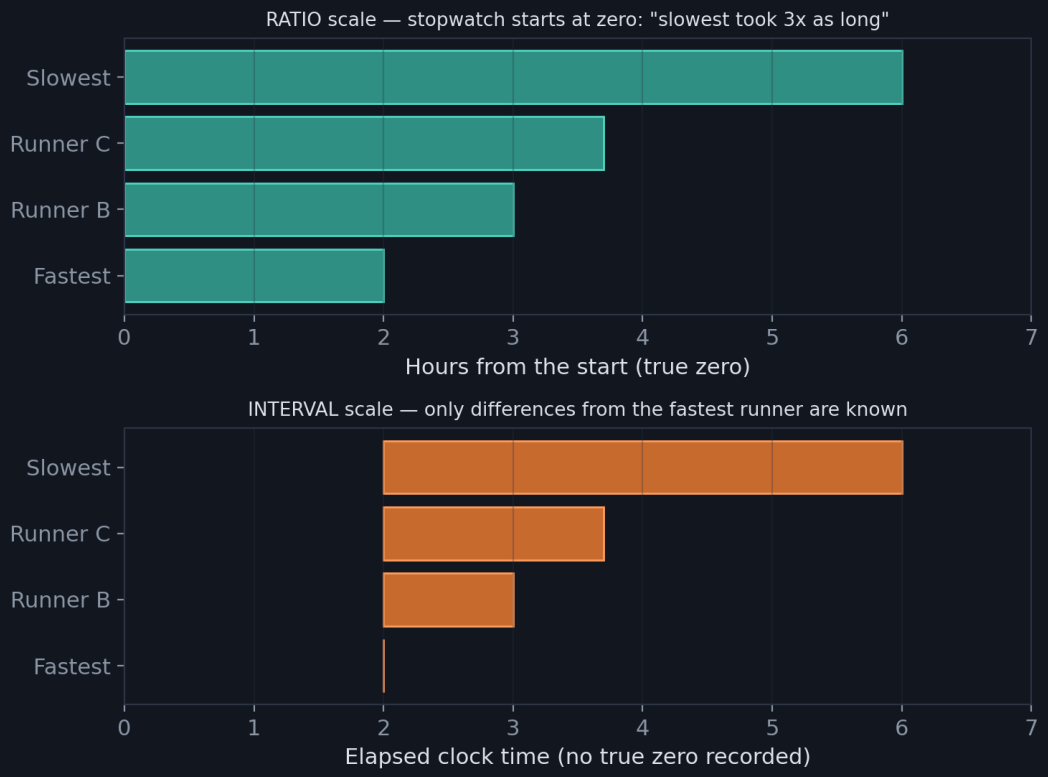


Figure 3.2 — Ratio scale (true zero, ratios are meaningful) vs. interval scale (equal intervals, but no true zero — only differences are meaningful).

	Equal intervals?	True zero point?	Ratios meaningful? (e.g., '2x as much')
Interval	Yes	No	No
Ratio	Yes	Yes	Yes

3.4 Worked Example: A School Commute Survey

Imagine surveying pupils in a school about how they get to school. Three questions, three different levels of measurement:

Question	Level	Why
“What mode of transportation do you use — bus, car, bicycle, walk?”	Nominal	Categories can be counted, but bus is not “higher” or “lower” than bicycle.
“How satisfied are you with your mode of transportation?” (very unsatisfied → very satisfied)	Ordinal	Responses can be ranked, but the exact gap between “satisfied” and “very satisfied” isn't quantifiable.
“How many minutes does it take you to get to school?”	Metric (ratio)	Exact numeric differences and averages are meaningful — you can calculate a true average commute time.

This example nicely summarizes the ladder of information: nominal data lets you categorize and count, but not order; ordinal data lets you rank, but not measure precise gaps; metric data lets you measure exact differences and use the full range of statistical tools.

3.5 Practice: Classify These Variables

Try to classify each variable below before checking the answer column.

Variable	Level	Why
US state of residence	Nominal	Labels/categories with no inherent order.
Product rating (1–5 stars)	Ordinal	Has order (5 > 1), but gaps between stars aren't guaranteed equal.
Religious confession	Nominal	Categories for classification only, no ranking.
CO2 emissions per year	Ratio	Zero means zero emissions; ratios like “twice as much” are meaningful.
Telephone number	Nominal	Numeric-looking, but purely an identifier — no quantitative meaning.
Patient care level (low/medium/high)	Ordinal	Ordered categories, but the size of the gap between levels is undefined.
Living space (square meters)	Ratio	Zero means no space; “twice as big” is meaningful.

Variable	Level	Why
Shop satisfaction (1–4 scale)	Ordinal	Ranked levels, but differences between levels aren't precisely quantified.

✓ Levels of measurement, remembered simply

- Nominal: name/label only — no order (e.g., colors, gender, brands).
- Ordinal: ordered, but unequal/unknown gaps (e.g., ratings, rankings, education level).
- Interval: equal gaps, no true zero (e.g., temperature in °C, calendar year).
- Ratio: equal gaps AND a true zero (e.g., weight, income, age, time elapsed).

Where to Go From Here

This tutorial covered the essentials that almost every statistical analysis builds on:

- What statistics is, and the difference between descriptive and inferential statistics.
- The core descriptive tools: measures of central tendency, measures of dispersion, frequency/contingency tables, and charts.
- The logic of hypothesis testing: null vs. alternative hypotheses, the p-value, statistical significance, and Type I / Type II errors.
- The three variants of the t-test — one-sample, independent-samples, and paired-samples — including their assumptions, hypotheses, and how to move from a t-value to a decision.
- The four levels of measurement — nominal, ordinal, interval, and ratio — and why they determine which analyses and charts are valid.

From here, the natural next steps are analysis of variance (ANOVA) for comparing more than two groups, correlation and regression analysis for relationships between variables, and cluster analysis for finding structure in data without predefined groups.

This document is a written tutorial adapted from a full video course on statistics, expanded with additional worked examples and original visualizations for clarity. This edition has been reformatted with a dark color palette for comfortable low-light reading.